

Rebekah Mugenyi

Wakiso, UG

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EDUCATION

Uganda Christian University, Mukono, UG

Bachelors of Sciences, **2024 - Ongoing**

Major: Electronics and Communication Engineering

Honours College Member

Language of Instruction: English

Lidri School International, Entebbe, UG

ICCE Certification

A-levels (Math Bias)

Physics, Chemistry, Math, Biblical Studies

CAREER EXPERIENCE

Lwera Semiconductors Intern Namanve, UG, **July 2026 - present**

- **Learning PCB design** PCB design fundamentals through KiCad, covering schematic capture and board layout/routing for electronic circuits
- **Skill: soldering** Practicing hand-soldering techniques across both surface-mount (SMD) and through-hole (THT) components, building precision and dexterity for electronics work

Teaching Assistant at Lidri School International Entebbe, UG, **May 2025 - August 2025**

- **Taught English language** to middle school students, delivering structured lessons focused on reading and grammar
- **Monitored and supported elementary school** pupils' academic progress and classroom engagement, providing guidance and feedback to promote steady learning development

Early Childhood Educator at Lidri School International, Entebbe, UG, **January 2024 - April 2024**

- **Instructed the ABC (reading class) using the Accelerated Christian Education (ACE) curriculum**, delivering individualized learning experiences aligned with each student's academic level and pace..
- **Designed and implemented extracurricular activity plans** tailored to address students' unique interests and areas for improvement, fostering holistic development beyond the core curriculum
- **Monitored and evaluated student progress regularly**, adapting instructional strategies and support mechanisms to ensure measurable academic and personal growth

Librarian at Lidri School International, Entebbe, UG, **May 2023 - April 2024**

- **Designed and implemented a comprehensive system of library policies, procedures, and guidelines**, resulting in improved efficiency and effectiveness of library operations.
- **Developed a structured cataloging system** that organized available books by genre and age-appropriate categories, enhancing accessibility for students and staff.

- **Established a mandatory reading program** requiring each student to read a minimum of four books per academic term, incorporating a tiered reward system tailored to each grade level to foster engagement and motivation.
- **Created age-appropriate book summary templates** aimed at cultivating foundational writing skills and encouraging literary expression in students under the age of 12.

Freelance Piano Tutor, Entebbe, UG, **January 2021** - Present

- **Instructing Classical Music**:, teaching students to play classical music as well as music theory
- **Instruction in Contemporary Music** : instruction on the application of the Nashville Number System

Programming Tutor, Kampala, UG, **2016**

- **Taught children programming**, using the SCRATCH programming language

PROJECTS

Hexapod Robot

Featured on NTV (Uganda's national broadcaster) for showcasing the hexapod's capabilities to a national audience.

Presented under Department of Computing, Uganda Christian University at National Council for Higher Education

(NCHE) Exhibition - 2nd Place (2025) | 1st Place (2026)

- Designed and built a six-legged autonomous hexapod robot using an Arduino Mega and MG996R servos to explore the potential of AI and machine learning integration in educational robotics.
- Engineered the robot's mechanical structure and programmed gait algorithms to achieve coordinated leg movements, including straight-line walking and rotational turning.
- Integrated YOLOv8n ONNX-based person detection via a Raspberry Pi, enabling real-time computer vision for autonomous response to detected objects and people
- Focused on creating a modular, scalable design suitable for AI-based upgrades such as adaptive terrain response.

Prototype Industrial Line-Following Rover

Uganda Industrial Research Institute (UIRI), 2025

- Designed and built a prototype industrial rover intended to automate inter-lab material transport at UIRI, following a defined line path between laboratory stations.
- Implemented line-following algorithms using IR sensors for reliable path detection and accurate navigation across facility floors.
- Integrated mobile application control, enabling operators to remotely command the rover.

SKILLS

- **Programming**: C++ for embedded systems programming and hardware interfacing, Python, Verilog
- **AutoCAD**: 2D Design, creating technical drawings and schematics
- **SolidWorks**: 3D Modeling, designing mechanical structures and prototypes in robotics and electronics enclosures
- **Typesetting LaTeX**: technical documents, academic reports, structured formatting

LEADERSHIP

President of the Robotics & AI Chapter, Uganda Christian University,

January 2025 - Present

- Leading the university Robotics Club, organising technical workshops, outreaches and project showcases

Residential Assistant (RA), Uganda Christian University,**January 2026 - April 2026**

- Coordinated accommodation logistics and served as a point of contact for visiting university delegations, ensuring smooth and professional experience for guests at the residential facility.
- Provided practical support to resident students, addressing welfare concerns and connecting students with appropriate university resources.
- Upheld and enforced residence policies and community standards, fostering a safe, respectful, and conducive living environment.
- Handled administrative duties including incident reporting, room allocation assistance, and liaising with university facilities staff on maintenance and operational matters.

LICENSES & CERTIFICATIONS

Workshop Practice:**January 2025 - April 2025**

Embedded Systems, 3D Modelling and Simulation, Domestic Electrical Installation, Computer Hardware Repair and Maintenance, Writing and Scholarly Skills, Printed Circuit Board Design and Simulation, Technopreneurship & Project Management, Local Area Networking (Certified by Uganda Christian University)

Uganda Industrial Research Institute (UIRI):**2025**

Completed technical training in electrical installations and embedded systems.

Introduction to Cyber Security: Cisco Certification**2024****Piano Grade 8:** Piano Performance Examination, ABRSM, *Distinction***2023****Material Science:10 things Every Engineer Should Know:** UC Davis online course**2023****International Computers Drivers License (ICDL):** APTECH**2019****Music Theory Grade 6:** Associated Board of the Royal Schools of Music Certification**2018****AWARDS**

- **Excellence Award** (Teacher's category): Consistent delivery of high quality work
- **Innovation Award** (Teacher's category): Implementation of creative teaching methods to maximize retention and creativity
- **Future Relevance Award** (Student category): Awarded during A-levels at Lidri School International
- **Excellence Award** (Student category): Awarded during A-levels at Lidri School International
- **Fearless Award (Youth Award):** For exemplary service in children's ministry

VOLUNTEERING EXPERIENCE

ROTOM (Reach One Touch One Ministries).

- School Management Committee (SMC) Member
- Grant Writing Volunteer: Collaborated on developing a grant proposal for the *Play Learn Thrive* initiative

**August 2025 - Present
April 2025****Watoto Church Morph Camp:**

A camp that hosts 1000 - 2000 teenagers for a week every year

- Camp 'Echo' Facilitator
- Camp 'Mirror Me' Facilitator

**January 2026
January 2025****Watoto Church Ministries:** Pianist and saxophonist for Sunday service**2022 - Present**

- Pianist and saxophonist for Sunday service
(Youth Ministry: Watoto Morph, 2020–2022; Main congregation, 2022–Present)

Mavuno Church - Children's Church Ministry:**2016**

- Musician serving in children's ministry

LANGUAGES

Language: English: Full professional proficiency